

[Edit](#)**Guided**

Craft Better Prompts in Azure OpenAI Studio

Challenge Overview

Understand the scenario

You are a <GlobalPowerUser> for Hexelo, an organization that needs assistance creating and improving AI prompts for upcoming projects. In this Challenge Lab you will create a new OpenAI service, and then you will create a new deployment. Next, you will improve basic prompts, and then you will create Chain-of-Thought prompting (for reasoning tasks). Finally, you will create prompting for Structured Output.

Navigating the Challenge Lab

🔗 ► Understanding a Cloud Slice

🔗 ► Quick tips for navigating the Challenge Lab instructions.

[New to Challenge Labs? Click here to learn more.](#)

Create a new OpenAI service

Hints Enabled

No Yes

 Due to high volume of Azure OpenAI service and resource requests within all regions of Azure, deployment of resources and select API endpoints are experiencing longer than anticipated deployment times. 

- Sign into the virtual machine as **Admin** using `T` `Passw0rd!` as the password.
- Open **Microsoft Edge**, go to `T` `https://portal.azure.com`, sign in to the Microsoft Azure portal as `T` `{USERNAME}` using `T` `{PASSWORD}` as the password, and then dismiss all prompts.

 Expand this hint for guidance on signing into the Microsoft Azure portal. 

- On the Windows taskbar, in Search, enter **Edge**, and then select **Microsoft Edge** to open a new browser.
- In the Address bar, enter `T` `https://portal.azure.com`, and then select **Enter**.
- In the Sign in dialog, enter `T` `{USERNAME}`, and then select **Next**.
- In Enter password, enter `T` `{PASSWORD}` and then select **Sign in**.
- Dismiss all prompts.

 Want to learn more? Review the documentation on [the Azure portal](#).

 If you receive a pop up window titled **Save password**: 

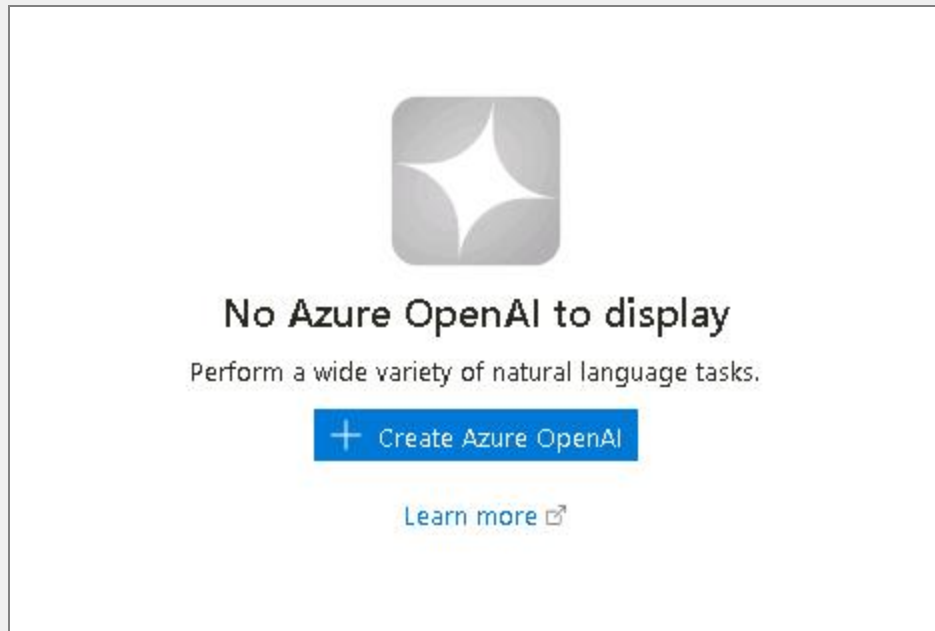
 Want to learn more? Review the documentation on [the Azure portal](#).

- Create a new `T` `Azure OpenAI` service by using the values from the following table. For any Properties not listed, leave those properties at their default values.

Property	Value
Subscription	Choose default
Resource group	{RESOURCE_GROUP_NAME}
Region	East US
Name	T OpenAI{LAB_INSTANCE_ID}
Pricing tier	Standard S0

💡 Expand this hint for guidance on creating an OpenAI resource in Azure.

- On the Azure home page, select **Create a resource**.
- In the Search services and marketplace field, enter **T Azure OpenAI**, and then press **Enter**.
- In the search results, in the Azure OpenAI box, select **+ Create Azure OpenAI**.



- On the Basics tab of the Create Azure OpenAI window, in Subscription, verify that the default subscription is selected.
- In Resource group, select the drop-down, and then select **{RESOURCE_GROUP_NAME}**.
- Confirm the Region field is set to **East US**. If not, select the drop-down arrow, and then select **East US**.
- In the Name field, enter **T OpenAI{LAB_INSTANCE_ID}**.
- In the Pricing tier field, select the drop-down arrow, select **Standard S0**, and then select **Next**.
- On the Network tab, confirm that **All networks, including the internet, can access this resource** is selected, and then select **Next**.
- Leave all the defaults on the Tags tab, and then select **Next**.
- On the Review + submit tab, select **Create**.

💡 Want to learn more? Review the documentation on [creating a new Azure Open AI service](#).

- Wait for the resource to be created, select **Go to resource**, and then select, **OpenAI{LAB_INSTANCE_ID}**.



It will take a couple of minutes for the resource to be created.

- Select **Explore Azure AI Foundry portal**, and then leave the Azure OpenAI Studio page open for the next activity.



Expand this hint for guidance on opening Azure AI Foundry portal.



- On the OpenAI{LAB_INSTANCE_ID} Azure OpenAI Overview page, on the get started tab, select **Explore Azure AI Foundry portal**.



Explore and deploy

Explore and deploy the generative AI models, craft unique prompts for your use cases, and fine-tune select models.

Explore Azure AI Foundry portal



Azure OpenAI Studio enables you to unleash the power of OpenAI's language models for text generation, question answering, and more. You can access pre-built models or train your own. You can also build intelligent applications quickly and securely.

Check your work

Verify

Start with basic prompts and improve them

Hints Enabled

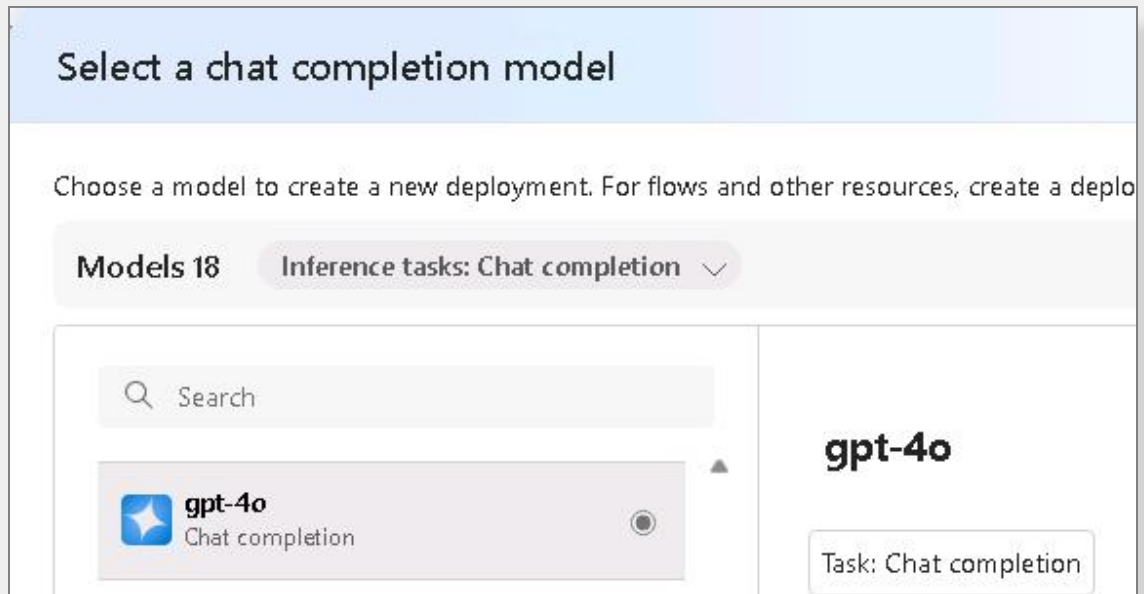
No Yes

- Create a new Deployment by using the values from the following table. For any Properties not listed, leave those properties at their default values.

Property	Value
Deploy model	gpt-4o
Model version	Auto-update to default
Deployment name	T AzureAI-Deployment{LAB_INSTANCE_ID}

💡 Expand this hint for guidance on creating a new deployment.

- On the Chat playground page, in Setup, select **Create a new deployment**.
- In the Select a chat completion model window, search for and select **gpt-4o**, and then select **Confirm**.



- In Deployment name, enter `T AzureAI-Deployment{LAB_INSTANCE_ID}`, and then select **Deploy**.

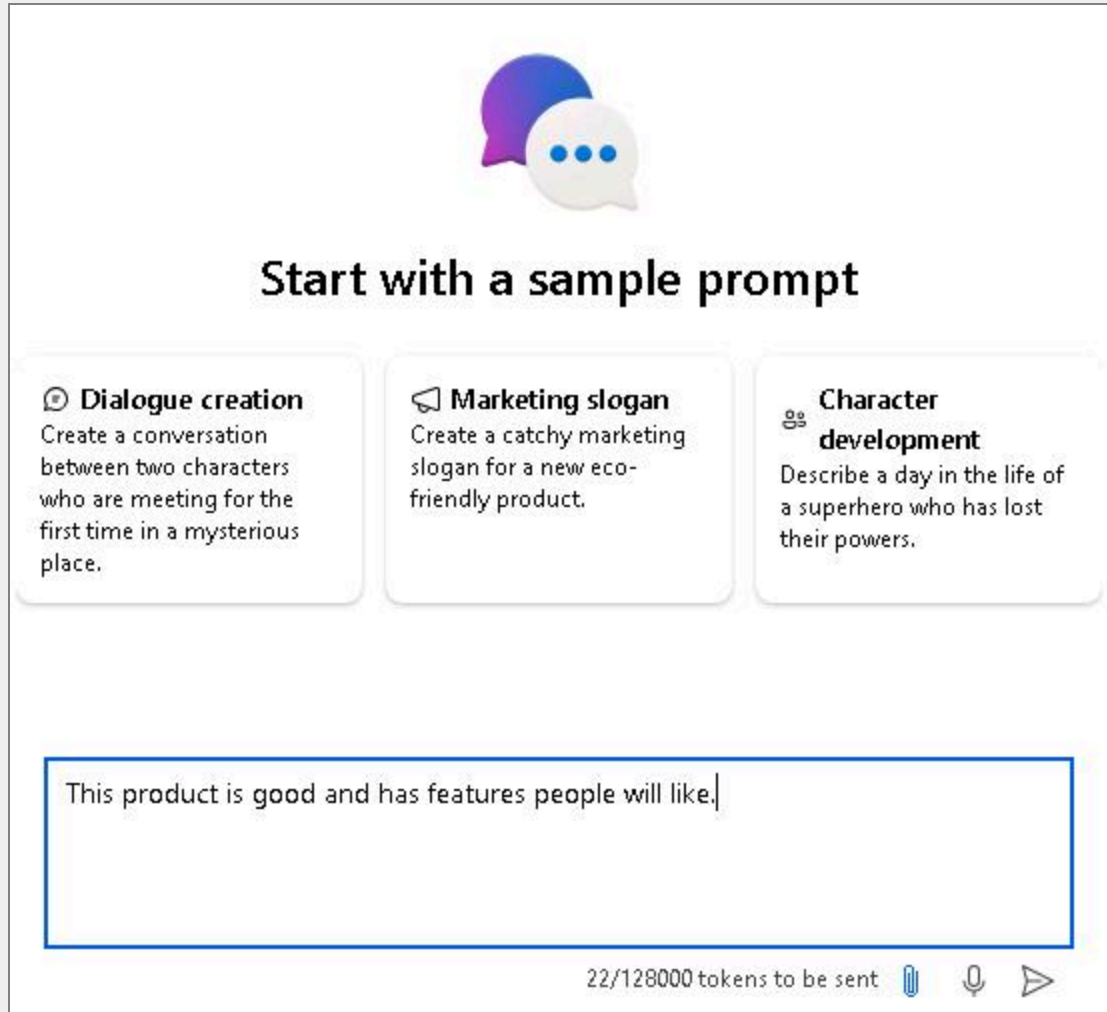
💡 Want to learn more? Review the documentation on [creating a new deployment](#).

- Enter the following prompt and then review the answer.

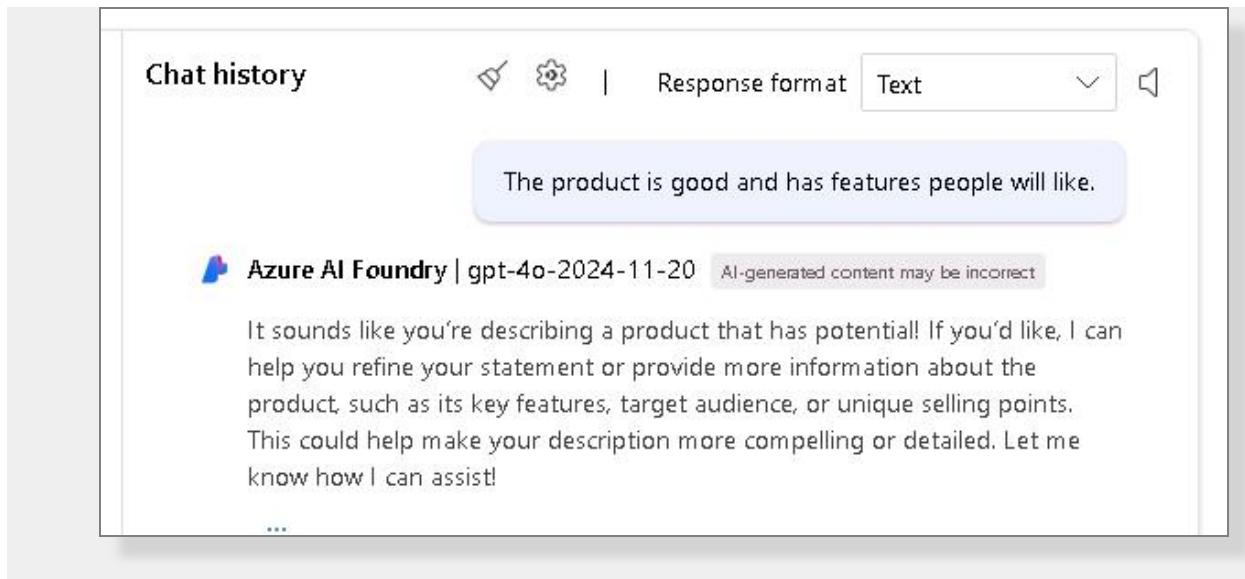
`T` The product is good and has features people will like.

💡 Expand this hint for guidance on entering a query in the chat playground. ^

- On the Azure AI Foundry| Azure OpenAI Chat playground page, in the *Type user query here* box, enter **T** The product is good and has features people will like., and then select the submit icon.



- Select **Send**, and then review Azure AI Foundry's response.

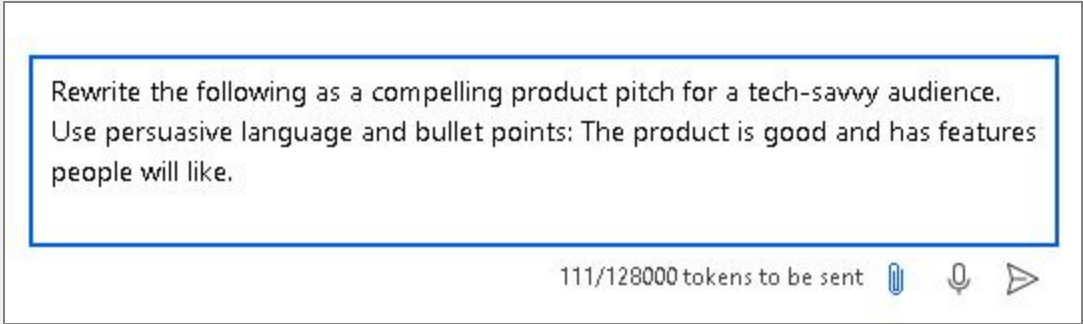


- Improve the previous prompt by using the following:

T Rewrite the following as a compelling product pitch for a tech-savvy audience. Use persuasive language and bullet points: The product is good and has features people will like.

 Expand this hint for guidance on improving a prompt.

- On the Azure AI Foundry| Azure OpenAI Chat playground page, in the *Type user query here* box, enter **T** Rewrite the following as a compelling product pitch for a tech-savvy audience. Use persuasive language and bullet points: The product is good and has features people will like., and then select the submit icon.

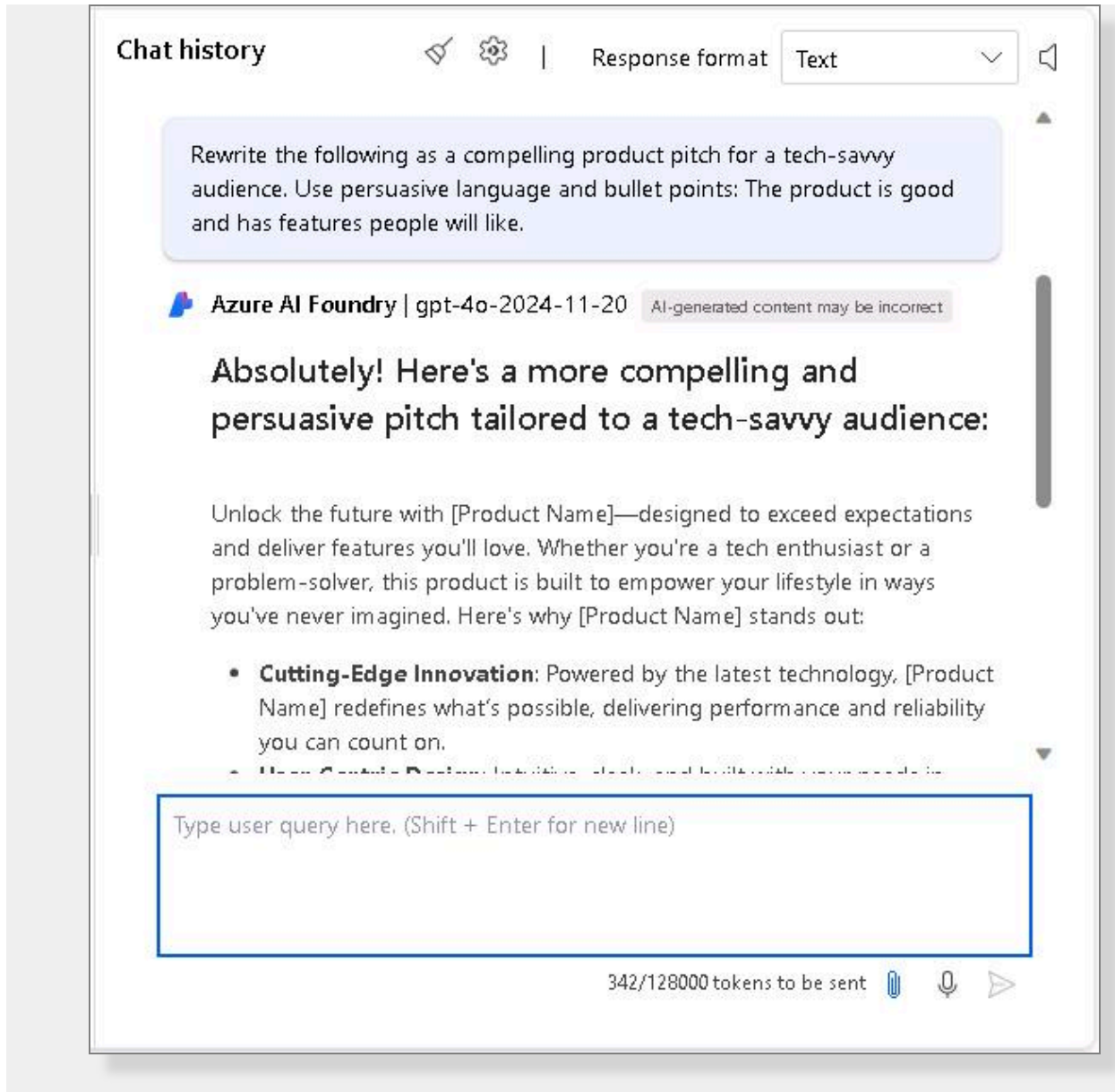


Rewrite the following as a compelling product pitch for a tech-savvy audience.
Use persuasive language and bullet points: The product is good and has features
people will like.

111/128000 tokens to be sent



- Select **Send**, and then review Azure AI Foundry's response.



- Enter the name in brackets into the following text box:

Improved Prompt

- Summarize the following text by entering **T** Summarize this:, and the following text, and then review the answer.

T Expand to view the Example text.

T In Q1 2025, our company achieved a 13.2% increase in revenue compared to the previous quarter, reaching \$18.6 million in total revenue. This growth was largely driven by a highly successful holiday season marketing campaign, which led to a 22% increase in new customer acquisitions. Additionally, improvements in our customer retention program resulted in a 15% increase in repeat purchases year-over-year. The expansion into European markets

contributed \$3.2 million to the quarterly revenue, led by strong adoption in Germany and the Netherlands. Operational costs rose by 5.8%, primarily due to increased shipping costs and temporary seasonal labor. Despite this, our gross margin improved by 2.5%, reaching 47.9%, due to better supplier negotiations and increased average order value. Customer satisfaction, as measured by Net Promoter Score (NPS), rose from 68 to 74, indicating improved service delivery and fulfillment times. Our mobile app saw a 34% increase in active users, and over 40% of total purchases were made through mobile platforms for the first time. Key challenges included supply chain delays for certain SKUs and increased competition in the apparel segment, which affected conversion rates in that category. The product team is addressing this with localized fulfillment centers and targeted merchandising strategies in Q2.

 Expand this hint for guidance on summarizing text.

- On the Azure AI Foundry| Azure OpenAI Chat playground page, in the *Type user query here* box, enter **T Summarize this;**, and then enter the Example text from above.

Summarize this: In Q1 2025, our company achieved a 13.2% increase in revenue compared to the previous quarter, reaching \$18.6 million in total revenue. This growth was largely driven by a highly successful holiday season marketing campaign, which led to a 22% increase in new customer acquisitions. Additionally, improvements in our customer retention program resulted in a 15% increase in repeat purchases year-over-year. The expansion into European markets contributed \$3.2 million to the quarterly revenue, led by strong adoption in Germany and the Netherlands. Operational costs rose by 5.8%, primarily due to increased shipping costs and temporary seasonal labor. Despite this, our gross margin improved by 2.5%, reaching 47.9%, due to better supplier negotiations and increased average order value. Customer satisfaction, as measured by Net Promoter Score (NPS), rose from 68 to 74, indicating improved service delivery and fulfillment times. Our mobile app saw

- Select **Send**, and then review Azure AI Foundry's response.

Chat history



Response format

Text



Azure AI Foundry | gpt-4o-2024-11-20

AI-generated content may be incorrect

- Revenue Growth & Key Drivers:** In Q1 2025, revenue grew 13.2% quarter-over-quarter to \$18.6M, fueled by a successful holiday marketing campaign (+22% new customer acquisitions), a 15% year-over-year increase in repeat purchases, and \$3.2M in contributions from European market expansion (led by Germany and the Netherlands).
- Profitability & Metrics:** Despite a 5.8% rise in operational costs due to shipping and seasonal labor, gross margin improved by 2.5% to 47.9%, driven by better supplier negotiations and higher average order values. Customer satisfaction rose, with NPS climbing from 68 to 74, and mobile platforms accounted for over 40% of purchases for the first time, supported by a 34% increase in app active users.
- Challenges & Next Steps:** Supply chain delays and increased competition in the apparel segment impacted conversion rates for specific SKUs. The product team is addressing these challenges in Q2 through localized fulfillment centers and targeted merchandising strategies.

- Improve the previous prompt by using the following:

T Summarize the previous report in 5 clear bullet points, focusing on growth drivers and metrics. 5 sentence maximum:

 Expand this hint for guidance on improving a prompt. 

- On the Azure AI Foundry| Azure OpenAI Chat playground page, in the *Type user query here* box, enter **T** Summarize the previous report in 5 clear bullet points, focusing on growth drivers and metrics. 5 sentence maximum:.

Summarize the previous report in 5 clear bullet points, focusing on growth drivers and metrics. 5 sentence maximum:

776/128000 tokens to be sent   

- Select **Send**, and then review Azure AI Foundry's response.



Azure AI Foundry | gpt-4o-2024-11-20

AI-generated content may be incorrect

- **Revenue Growth:** Revenue increased by 13.2% in Q1 2025 to \$18.6M, driven by a 22% rise in new customer acquisitions, a 15% year-over-year increase in repeat purchases, and \$3.2M from European market expansion, led by Germany and the Netherlands.
- **Profitability:** Gross margin improved by 2.5% to 47.9% despite a 5.8% rise in operational costs due to shipping and seasonal labor.
- **Customer Engagement:** NPS rose from 68 to 74, reflecting improved service delivery, while mobile platforms accounted for over 40% of purchases, supported by a 34% increase in app active users.
- **Key Drivers:** Strong holiday marketing campaign, better supplier negotiations, and increased average order value significantly contributed to growth.
- **Challenges & Actions:** Supply chain delays and heightened apparel competition impacted conversion rates, with plans for Q2 focused on localized fulfillment centers and targeted merchandising strategies.

- Enter the words in bold for the first bullet into the following text box:

Improved Summary

- Enter the following prompt, and then review the answer.

T Tell me about AI in schools.

 Expand this hint for guidance on entering a query in the chat playground. 

- On the Azure AI Foundry| Azure OpenAI Chat playground page, in the *Type user query here* box, enter T Tell me about AI in schools., and then select the submit icon.

Tell me about AI in schools.

881/128000 tokens to be sent   

- Select **Send**, and then review Azure AI Foundry's response.



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AI-generated content may be incorrect

AI in schools is transforming education by enhancing learning experiences, streamlining administrative tasks, and providing personalized support for students and educators. Here's an overview of how AI is being integrated into schools:

Applications in Education

1. **Personalized Learning:** AI-powered tools can analyze student performance and learning styles to offer tailored lesson plans, adaptive learning platforms, and individualized feedback, helping students learn at their own pace.
2. **Tutoring and Support:** AI chatbots and virtual tutors provide instant assistance with homework, answering questions, and explaining concepts, making education accessible outside of classroom hours.
3. **Administrative Efficiency:** AI automates repetitive tasks such as

- Improve the Format and Tone of a prompt by entering the following, and then review the answer.

T You are an educational technology consultant. Write a professional email explaining the benefits of using AI tools in high school classrooms to a school district superintendent.

💡 Expand this hint for guidance on improving the format and tone of a prompt. ^

- On the Azure AI Foundry| Azure OpenAI Chat playground page, in the *Type user query here* box, enter **T** You are an educational technology consultant. Write a professional email explaining the benefits of using AI tools in high school classrooms to a school district superintendent., and then select the submit icon.

You are an educational technology consultant. Write a professional email explaining the benefits of using AI tools in high school classrooms to a school district superintendent.

1252/128000 tokens to be sent

- Select **Send**, and then review Azure AI Foundry's response.



Azure AI Foundry | gpt-4o-2024-11-20

AI-generated content may be incorrect

Subject: Unlocking the Potential of AI Tools in High School Classrooms

Dear [Superintendent's Name],

I hope this email finds you well. As an educational technology consultant, I am passionate about supporting schools in adopting innovative tools that enhance student learning and empower educators. I wanted to share insights into how AI tools can transform high school classrooms in your district by fostering personalized learning, improving engagement, and streamlining administrative processes.

AI-powered platforms are revolutionizing education by tailoring instruction to individual student needs. Adaptive learning systems analyze student performance in real time, providing personalized lesson plans and targeted

- Enter the first bolded word of the email into the following text box:

Format Tone Prompt

- Enter the following Reasoning prompt, and then review the answer.

T You are a math tutor. A student asks: "Is 153 divisible by 3?" Think through the solution step-by-step before giving the final answer.

 Expand this hint for guidance on creating a Reasoning prompt.

- On the Azure AI Foundry| Azure OpenAI Chat playground page, in the *Type user query here* box, enter **Tell me about AI in schools.**, and then select the submit icon.

You are a math tutor. A student asks: "Is 153 divisible by 3?" Think through the solution step-by-step before giving the final answer.

1484/128000 tokens to be sent



- Select **Send**, and then review Azure AI Foundry's response.



Azure AI Foundry | gpt-4o-2024-11-20

AI-generated content may be incorrect

To determine if 153 is divisible by 3, we can use the divisibility rule for 3, which states that a number is divisible by 3 if the sum of its digits is divisible by 3. Here's the step-by-step process:

1. Add the digits of 153:

The digits of 153 are 1, 5, and 3.
($1 + 5 + 3 = 9$).

2. Check if the sum (9) is divisible by 3:

Since 9 is divisible by 3 (because ($9 \div 3 = 3$) with no remainder), this means 153 is also divisible by 3.

3. Verify with direct division:

Divide 153 by 3:
($153 \div 3 = 51$).
There is no remainder, confirming that 153 is divisible by 3.

Final Answer: Yes, 153 is divisible by 3.

...

- Enter the Final answer into the following text box:

Reasoning

- Leave the Azure AI Studio open for the next activity.

Check your work

Verify

Prompting for Structured Output

Hints Enabled

No Yes

- Extract structured insights from unstructured text and return as JSON by using the following:

T Extract the following meeting transcript into structured JSON with keys: `action_items` (array), `decisions` (array), `attendees` (array of names), `next_meeting_date` (ISO 8601 format). Transcript: Thanks everyone for joining today. Alice and Bob will follow up on the pricing proposal by next Friday. We agreed to delay the product launch until June 15. Charles noted that marketing will need 3 more weeks. Present: Alice, Bob, Charles, Dana. Let's meet again on June 12 at 10am.

 Expand this hint for guidance on creating a Reasoning prompt.

- On the Azure AI Foundry| Azure OpenAI Chat playground page, in the *Type user query here* box, enter the example prompt from above, and then select the submit icon.

Extract the following meeting transcript into structured JSON with keys: action_items (array), decisions (array), attendees (array of names), next_meeting_date (ISO 8601 format). Transcript: Thanks everyone for joining today. Alice and Bob will follow up on the pricing proposal by next Friday. We agreed to delay the product launch until June 15. Charles noted that marketing will need 3 more weeks. Present: Alice, Bob, Charles, Dana. Let's meet again on June 12 at 10am.

353/128000 tokens to be sent   

- Select **Send**, and then review Azure AI Foundry's response.



Azure AI Foundry | gpt-4o-2024-11-20

AI-generated content may be incorrect

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3. Verify with direct division:

Divide 153 by 3:
($153 \div 3 = 51$).
There is no remainder, confirming that 153 is divisible by 3.

Final Answer: Yes, 153 is divisible by 3.

...

- Enter the **next_meeting_date** in JSON format into the following text box:

JSON Format

- Create a prompt to extract text to CSV Format.

T From the following meeting transcript, extract the key action items, decisions, attendees, and next meeting date. Output the results in CSV format with columns: Action Item, Decision, Attendee, Next Meeting Date. Transcript: Thanks everyone for joining today. Alice and Bob will follow up on the pricing proposal by next Friday. We agreed to delay the product launch until June 15. Charles noted that marketing will need 3 more weeks. Present: Alice, Bob, Charles, Dana. Let's meet again on June 12 at 10am.

 **Expand this hint for guidance on extracting a prompt to CSV format.** 

- On the Azure AI Foundry| Azure OpenAI Chat playground page, in the *Type user query here* box, enter the example prompt from above, and then select the submit icon.

From the following meeting transcript, extract the key action items, decisions, attendees, and next meeting date. Output the results in CSV format with columns: Action Item, Decision, Attendee, Next Meeting Date. Transcript: Thanks everyone for joining today. Alice and Bob will follow up on the pricing proposal by next Friday. We agreed to delay the product launch until June 15. Charles noted that marketing will need 3 more weeks. Present: Alice, Bob, Charles, Dana. Let's meet again on June 12 at 10am.

437/128000 tokens to be sent



- Select **Send**, and then review Azure AI Foundry's response.



Azure AI Foundry | gpt-4o-2024-11-20

AI-generated content may be incorrect

Here's the extracted information from the transcript in CSV format:



```
Action Item,Decision,Attendee,Next Meeting Date
Alice and Bob will follow up on the pricing proposal by
next Friday,,Alice,Bob,2023-06-12T10:00:00
,Delay the product launch until June 15,,Charles,2023-
06-12T10:00:00
,Marketing will need 3 more weeks,,Dana,2023-06-
12T10:00:00
```

- Enter the result in CSV format into the following text box:

CSV Format

--

- Create a prompt to extract text to Markdown format.

T Extract the following meeting transcript into Markdown formatted summary. Use the following structure: **Action Items, Decisions, Attendees, Next Meeting Date**. Transcript: Thanks everyone for joining today. Alice and Bob will follow up on the pricing proposal by next Friday. We agreed to delay the product launch until June 15. Charles noted that marketing will need 3 more weeks. Present: Alice, Bob, Charles, Dana. Let's meet again on June 12 at 10am.

💡 Expand this hint for guidance on extracting a prompt to a markdown format. ^

- On the Azure AI Foundry| Azure OpenAI Chat playground page, in the *Type user query here* box, enter the example prompt from above, and then select the submit icon.

Let me know if you need

Extract the following meeting transcript into a Markdown-formatted summary. Use the following structure: Action Items, Decisions, Attendees, Next Meeting Date. Transcript: Thanks everyone for joining today. Alice and Bob will follow up on the pricing proposal by next Friday. We agreed to delay the product launch until June 15. Charles noted that marketing will need 3 more weeks. Present: Alice, Bob, Charles, Dana. Let's meet again on June 12 at 10am.

575/128000 tokens to be sent

- Select **Send**, and then review Azure AI Foundry's response.

 **Azure AI Foundry** | gpt-4o-2024-11-20 AI-generated content may be incorrect

Here's the Markdown-formatted summary of the meeting:

Action Items

- Alice and Bob will follow up on the pricing proposal by next Friday.

Decisions

- The product launch will be delayed until June 15.
- Marketing will need 3 more weeks.

Attendees

- Alice
- Bob
- Charles
- Dana

- Enter **T** [View in Markdown](#), select **Send**, and then enter the result in Markdown format into the following text box:

Here is the Markdown-formatted summary of the meeting:



```
### Action Items
- Alice and Bob will follow up on the pricing proposal
  by next Friday.

### Decisions
- The product launch will be delayed until June 15.
- Marketing will need 3 more weeks.

### Attendees
- Alice
- Bob
- Charles
- Dana

### Next Meeting Date
- June 12, 2023, at 10:00 AM (ISO 8601: `2023-06-
```

Markdown Format



Want to learn more? Review the documentation on [using the Azure AI Studio Chat Playground](#).

Check your work

Verify

Summary

Congratulations, you have completed the **Craft Better Prompts in Azure OpenAI Studio** Challenge Lab.

You have accomplished the following:

- Created a new OpenAI service.
- Created a new deployment.
- Started with basic prompts and improve them.
- Created Chain-of-Thought prompting (for reasoning tasks).
- Created prompting for Structured Output.

Ending your lab

To ensure your lab is recorded as complete, select **Submit** or **End** below. Exiting [X] the lab will result in an incomplete status.

 Once you select **Submit** or **End**, you will not be able to return to this Challenge Lab.

Your feedback is important!

As you end your Challenge Lab, please take a few minutes to complete the short survey that will appear in the next window. Alternatively, you may provide your feedback directly to [Challenge Labs feedback](#).

Looking for your next Challenge Lab?

Below are recommended Challenge Labs to try next.

- Manage Azure Resource Deployment by Using an Azure Resource Manager Template [Guided]
- Manage Azure Resource Groups [Guided]
- Create an Azure Function App [Guided]
- Create an Azure Logic App [Guided]
- Configure an Azure Distributed Denial of Service Protection Plan [Guided]

- Manage Encryption by Using an Azure Key Vault [Guided]
- Configure Azure Role-Based Access Control [Guided]
- Configure an Azure Lock [Guided]
- Configure Monitoring by Using Azure Monitor [Guided]
- Deploy an Azure Virtual Machine [Guided]
- View Azure Service Health Options [Guided]
- Manage Microsoft Entra Users and Groups [Guided]
- Implement a Network Security Group [Guided]
- Azure Cost Management [Guided]

